

GAURAV GANDHI

Lead Data Scientist · Applied AI | GenAI & LLM Systems · Multi-Agent RAG · Fine-Tuning · ML Evaluation

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PROFESSIONAL SUMMARY

Lead Data Scientist heading a 5-person team within Uber's AI organization, responsible for the modeling roadmap, evaluation strategy, and technical review of document-intelligence and conversational-AI systems, including the verification platform that gates earner onboarding across all Uber regions. Shipped a document-understanding Transformer that runs in production at 95%+ field accuracy and saves over \$10M annually, and a vision quality gate that automates roughly 70% of document verification. Hired and onboarded two of the team's data scientists; works daily with engineering, legal, annotation, and regional leadership teams. Technical depth spans LLM fine-tuning (LoRA/QLoRA), multi-agent RAG, distributed training (Ray Train, DeepSpeed ZeRO-3), model calibration, and conformal prediction. Success metrics and evaluation gates are defined before development begins, and systems are monitored in production after release.

PROFESSIONAL EXPERIENCE

Uber Technologies — Uber AI | via Indium Software · Bengaluru, India

Jun 2024 – Present

Lead Data Scientist

Aug 2026 – Present

- Lead a 5-person data-science team across GenAI document-intelligence and conversational-AI workstreams, setting the modeling roadmap, evaluation strategy, and technical review standards; hired two of the team's data scientists through end-to-end interviews and onboarded both onto live production systems.
- Coordinate the document-verification platform that gates earner onboarding in every Uber region, working with engineering, legal, annotation, regression, and regional leadership teams on model updates and threshold rollouts.

Senior Data Scientist

May 2025 – Aug 2026

- Automated roughly 70% of earner document verification (thousands of documents daily) by designing a dual-threshold policy on a Vision Transformer: recall-calibrated auto-approval and precision-calibrated auto-rejection thresholds, tuned per document type and rejection reason (55 types, 9 reasons), so only the uncertain band reaches human review.
- Raised field-level extraction accuracy above 97% by building a parameter-efficient fine-tuning framework for Qwen (LoRA/QLoRA) with a layer-freezing strategy, instruction-tuning data design, and an automated field-accuracy evaluation harness; mentored partner teams on prompt and evaluator design.
- Delivered \$10M+ in annual cost savings by displacing manual document review: owned the multi-task training objective and extraction head of an encoder-decoder Transformer pretrained on 50M+ documents across 144 A100 GPUs (PyTorch, Ray Train, DeepSpeed ZeRO-3), shipped to production at 95%+ field accuracy.

Data Scientist

Jun 2024 – May 2025

- Cut analytics turnaround from hours to seconds by shipping a multi-agent copilot that combines hybrid RAG (dense + sparse retrieval), NL2SQL, and a schema knowledge graph — adopted by program managers across all business units.
- Drove a sustained 27% lift in user engagement, validated through A/B tests across releases, by replacing static collaborative filtering with a BERT-based session-aware sequence model that captures evolving user intent.

FedEx Express | Consumer Logistics & Supply Chain · Bengaluru (Remote)

Aug 2022 – Jun 2024

Decision Scientist

- Recovered lost enterprise shipping volume, surfaced through Average-Daily-Volume decline, by deploying an ensemble of Bayesian change-point detection, probabilistic clustering, and profitability scoring, and worked cross-functionally with sales and operations to shape regional supply strategy.
- Reduced operational inefficiencies by roughly 20% by upgrading ARIMA pipelines to SARIMA with Box-Jenkins tuning, Fourier seasonality terms, and automated anomaly filtering, feeding directly into sales and staffing plans.

Tata Consultancy Services (TCS) | Bengaluru (Remote)

Jul 2021 – Jul 2022

Data Engineer

- Enabled reliable, scalable analytics for a high-volume booking and transaction client by designing end-to-end GCP ETL pipelines and dashboards for campaign targeting, segmentation, and trend monitoring.

APPLIED AI PROJECTS

Independently built and deployed. Full portfolio: gaurav-gandhi.vercel.app

- **TriagleIQ** — ML issue-triage service (GCP Cloud Run) Live · Code — Reached 87–90% top-3 accuracy in routing GitHub issues to the right component by combining a temperature-calibrated classifier, semantic retrieval of similar solved issues (BGE + FAISS), and conformal prediction intervals for resolution time; LLM fabrication rate is enforced as a hard CI gate.
- **Style Maitri** — AI stylist for Indian occasion wear Live · Code — Achieved 93.8% intent-parsing accuracy (n=211) and NDCG@10 of 0.918 over a 52,000-item catalogue from 8 stores by building a LangGraph agent on hybrid retrieval (FAISS + BM25 with cross-encoder reranking) and a grounding layer that removes invented price, size, and fabric claims.
- **Samidha Reviews** — multilingual review intelligence Live · Code — Reached 78.6% accuracy on a 43-review CI-gated eval (threshold 76%) across English, Hindi, and Hinglish, by building a tiered LLM extraction service (Llama 3.3 70B, JSON mode) with authenticity scoring, prompt-injection defense, PII redaction, and multi-tenant isolation.
- **AgentGauge** — causal A/B harness for AI-agent tool use (PyPI) Live · Code — Measured a 13–29 percentage-point drop in agent task success caused by a single defective tool description, across three model families, using a pre-registered A/B design with fully deterministic replay and per-run cost accounting.
- **tracegauge** — cost and efficiency gauge for AI coding-agent sessions (PyPI) Live · Code — Scores agent sessions on token economy, trajectory quality, and deterministic waste detection, and gates cost regressions in CI using bootstrap confidence intervals rather than point estimates; 2 contributions landed in google/adk-python (#6681, #6939); 1 in keras-team/keras (#23420).

RESEARCH

Two first-author working papers (drafts, not yet submitted):

- On tool descriptions for AI agents: controlled experiments across three model families showing that more precise tool descriptions help agents choose between similar tools, but hurt performance on broader selection and retrieval tasks — description quality is not a single dial to turn up.
- On the statistics of agent benchmarks: shows that many agent evaluations are too noisy to detect the effects they report, and contributes a method for computing the smallest effect a benchmark can reliably detect, plus a screening protocol for measurement artifacts.

TECHNICAL SKILLS

- **Leadership & Delivery:** Technical Leadership, Modeling Roadmap Ownership, Hiring & Interviewing, Mentorship, Stakeholder Management, Cross-Functional Delivery, Evaluation Strategy, Technical Review
- **ML & Deep Learning:** Transformers (BERT, Llama, GPT), ViT, CLIP, Diffusion Models (SDXL, ControlNet), Two-Tower & Contrastive Learning, Recommender and Ranking Systems, Model Calibration, Conformal Prediction, A/B Testing & Experimentation, Causal Inference, Bayesian Inference, Time-Series Forecasting
- **LLMs & GenAI:** LoRA/QLoRA, PEFT, Instruction Tuning, Hard-Negative Mining, LLM-as-Judge Evaluation, RLHF/DPO, Guardrails & Grounding, Structured Output (JSON Mode), Prompt-Injection Defense, LLM APIs (OpenAI, Anthropic, Gemini, Groq)
- **RAG, Retrieval & Agents:** Hybrid Retrieval (Dense + BM25), Cross-Encoder Reranking, Vector Search (FAISS, pgvector), NL2SQL, Knowledge Graphs, Multi-Agent Systems, LangChain, LangGraph, MCP
- **Languages & Frameworks:** Python (Advanced), SQL, PySpark, PyTorch, Hugging Face Transformers, diffusers, sentence-transformers, scikit-learn, LightGBM, FastAPI, Pytest
- **Deployment & MLOps:** Docker, GCP (Cloud Run, Vertex AI, Workload Identity Federation), MLflow, GitHub Actions CI/CD, Dagster, Ray Train, DeepSpeed ZeRO-3, Production Monitoring

EDUCATION

Post Graduate Diploma in Data Science, Great Lakes Institute of Management · GPA 8.45/10	2022
B.Tech, Electronics & Communication Engineering, Manipal University Jaipur · GPA 8.05/10	2017 – 2021

CERTIFICATIONS

- Deep Learning, NLP, and Generative AI with LLMs / LLMOps Specializations · DeepLearning.AI
- Machine Learning with Python · IBM
- Google Cloud Professional Data Engineer · Google Cloud